GPhyT With and Without Attention Residuals: A Budgeted Paired Architecture Study

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Abstract

We compare GPhyT with Block Attention Residuals against GPhyT without Attention Residuals, reusing the released GPhyT backbone and Flash Linear Attention reference operator. Both models train from scratch with six paired seeds, identical shared-parameter initialization and sampled training examples, the same optimizer schedule, effective batch size 16, and 3,175 updates. Standard GPhyT retains its original additive residuals and transformer self-attention. Three-seed Full and uniform-routing variants provide exploratory controls. The selected convection and radiative-flow datasets use official trajectory splits, training-only normalization, and 128×64 resolution. The primary endpoint compares equally weighted dataset log ratios of twelve-step autoregressive NMSE. The total project limit is 24 GPU-hours. At this revision, 18/18 planned training runs are complete. Relative to GPhyT without Attention Residuals, Block reduces the primary error endpoint by 24.52% (95% CI [11.56, 35.58]%; exact paired $p = 0.03125$). The interval supports the prespecified minimum 5% improvement. Claims are restricted to this model scale, selected data, resolution, and training budget. All four trained variants nevertheless perform far worse than persistence on convection under the prespecified metric. Radiative forecasts improve in error but produce negative-density cells. The relative architecture result therefore does not establish a physically reliable surrogate or a general advantage of learned routing over fixed averaging. Additional post-hoc zero-update controls show that training increases convection error for Block, Full, and Uniform Block in every tested seed, while reducing radiative-flow error for all four architectures. These controls leave the preregistered trained-model comparison unchanged but further limit its interpretation.

1 Context, motivation, and question

GPhyT uses a shared transformer architecture for forecasting heterogeneous physical fields from a short observed context[1, 2]. Attention Residuals replace fixed residual accumulation with content-dependent aggregation across network depth[3, 4]. Their reported language-model results do not establish an advantage for physics forecasts. We test the narrower question: under matched initialization, data schedules, and optimizer-update counts, does Block Attention Residual routing reduce GPhyT's autoregressive prediction error on two selected simulation datasets?

The minimum practically meaningful reduction is prespecified as 5%. Evidence is conditional on the selected data, resolution, architecture, and compute budget. A confidence interval whose upper improvement bound is below 5% contradicts that practical hypothesis at the tested scale; an interval spanning the threshold is inconclusive. Failure to reject a zero-effect null is not evidence of equivalence. This distinction follows general recommendations to separate empirical evidence, speculation, and limitations[12, 13].

2 Background and related implementations

The Well provides simulations with explicit trajectory splits and physical field metadata[5]. We use its uniform-grid Rayleigh–Bénard variant and two-dimensional turbulent radiative layer. The uniform-grid correction matters: the original convection data included a nonuniform spatial coordinate, and the maintainers released a separately resampled uniform version[6].

We reuse the released GPhyT code at commit `df1527c`; this is an architecture study trained from scratch, not a reproduction of its broad foundation-model training. For depth aggregation, we vendor the MIT-licensed FLA `naive_attnres` implementation at commit `a400aff6`, preserving source and license[4]. Fast-AttnRes and Liger provide optimized implementations that introduce Triton-specific execution dependencies[9, 10]. We also inspected nktkt's standalone implementation[11]. Our source scheduling is specified directly below and tested against raw sublayer outputs, rather than inheriting a third-party transformer integration. The reviewed alternatives, complete revisions, source URLs, and byte hashes are in the public repository[16].

3 Method

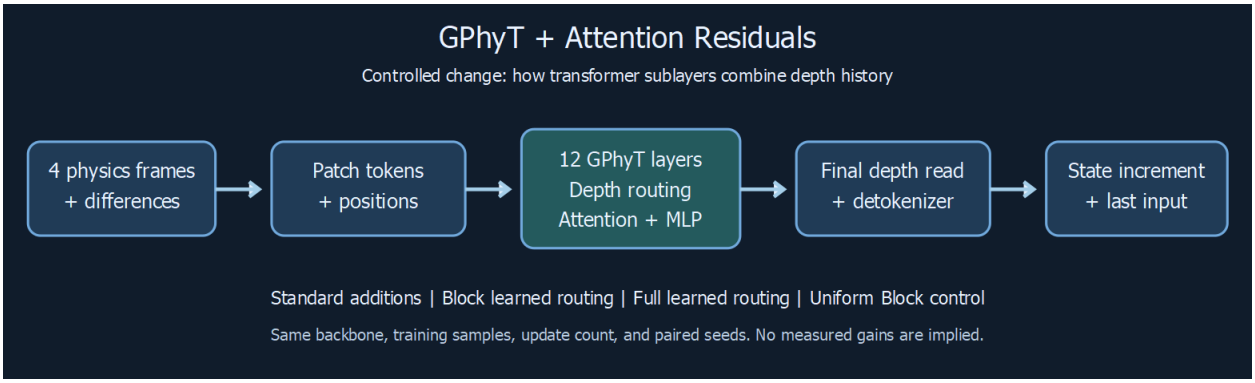


Figure 1: Controlled architectural change. All variants reuse the same upstream transformations and outer next-state update; only depth composition changes.

3.1 Controlled architecture change

The backbone has twelve transformer layers, hidden width 192, MLP width 768, three attention heads, absolute positional embeddings, and $1 \times 16 \times 16$ spatiotemporal patches. Four context frames at 128×64 give 128 tokens. We preserve the upstream tokenizer, positional embeddings, attention, MLPs, finite-difference features, and detokenizer. The upstream forward computes an increment for each context position, adds the original input, and returns the final position. Although the upstream class stores an integrator, this forward does not execute Runge–Kutta integration or use the `step_size` argument[2].

Write the tokenizer-plus-position output as e, and raw attention or MLP output at sublayer ℓ as $v_\ell = F_\ell(h_\ell)$. Each F_ℓ retains its original normalization. Given the causally available source bank

S_ℓ, our depth read is

$$k_{\ell i} = \gamma_\ell \odot s_i \left(d^{-1} \sum_{j=1}^{d} s_{ij}^2 + 10^{-6} \right)^{-1/2}, \tag{1}$$

$$a_{\ell i} = \frac{\exp(q_\ell^\top k_{\ell i})}{\sum_{s_r \in S_\ell} \exp(q_\ell^\top k_{\ell r})}, \qquad h_\ell = \sum_{s_i \in S_\ell} a_{\ell i} s_i. \tag{2}$$

These operations apply independently at each token. The query q_ℓ starts at zero and the key scale γ_ℓ at one[3, 4]. Mathematically, the initial read is an arithmetic mean, not a sum. Values are the unnormalized sources; only keys are RMS-normalized. There is no additional inverse-square-root attention scale in this operator.

Full routing stores $S_\ell = \{e, v_1, \ldots, v_{\ell-1}\}$. Block routing stores the embedding, completed sums of four consecutive sublayer outputs, and a nonempty partial sum for the current block. A completed sum appears exactly once: its partial slot is cleared at the boundary. Thus the 24 writers form six blocks. A separate final depth read includes all completed outputs before detokenization. The source bank resets for every forward call. Uniform Block uses exactly these sources with fixed equal weights and no learned routing parameters. Standard residuals call the unchanged upstream block forward. With five canonical channels, execution checks count 7,919,045 backbone parameters; learned routing adds $25 \times 2 \times 192 = 9,600$ parameters. This count is specific to our input/output configuration, rather than the parameter count of a differently configured published model.

3.2 Data and normalization

The selected datasets originate from Dedalus convection simulations and Athena++ radiative mixing-layer simulations, respectively; we credit their original numerical and physical work[7, 8] in addition to The Well distribution[5]. The convection selection fixes Prandtl number 1 and Rayleigh numbers 10^6, 10^8, and 10^{10}. The radiative selection includes all nine cooling settings. Official train, validation, and test trajectory membership is preserved. Source files, coordinates, parameter identities, SHA-256 hashes, and trajectory hashes are recorded before training. Exact full-trajectory duplication across splits causes preparation to fail. Shared initial-state hashes are disclosed separately because identical starting fields need not imply identical trajectories.

Each field is area-downsampled on CPU to 128×64 without transposing or reversing stored spatial axes. The canonical channel vector is pressure, density, buoyancy, velocity x, and velocity y. Buoyancy is not relabeled as temperature. Absent fields are zero, are masked out of optimization and error metrics, and remain zero in autoregressive feedback. Per-dataset, per-active-field means and standard deviations use all selected training trajectories only. Validation/test targets do not enter normalization.

The convection split contains 120 training, 15 validation, and 15 test trajectories; the corresponding radiative counts are 72, nine, and nine. Original grids are 512×128 and 128×384, with 200 and 101 frames per trajectory. No exact full-trajectory hash crosses splits. There are 33 distinct cross-split first-frame hash groups (25 convection, eight radiative), all spanning different parameter settings; every held-out trajectory shares an initial state with training under another setting. No first-four-frame prepared context is exactly duplicated across splits. This context check is not an exhaustive search over all possible windows. The public manifest and separate audit record all member identities[16].

Training samples are drawn uniformly by dataset, then parameter setting, trajectory, stride in $\{1, \ldots, 8\}$, and a valid start index. Four strided context frames predict the fifth. The model

receives no explicit parameter label or stride value. Validation monitoring uses a separate fixed sampling generator and cannot advance the training sequence. Evaluation uses stride one and three prespecified start fractions, 0, 1/2, and 1, of the valid origin range for four context plus 24 forecast frames.

3.3 Optimization and compute allocation

Under BF16 autocast, the upstream learned FP32 positional embedding promotes the residual stream to FP32. We promote raw sublayer outputs to that same stream dtype before depth-block summation, preserving the standard path's accumulation precision. The attention and MLP matrix operations retain BF16 autocast; this is mixed precision rather than a fully BF16 residual accumulator. Routing also executes inside autocast: FLA's explicit FP32 casts do not prevent its eligible contractions from being autocast[15]. The original protocol incorrectly described routing as entirely FP32. This discrepancy was disclosed during training; the frozen training computation was not changed. CPU BF16 regression checks reproduce the distinction; they are not measurements of CUDA rounding error. Standard and Block variants share seeds 0–5; Full and Uniform Block use seeds 0–2. Backbone initialization is identical within each seed across variants. Data RNGs are independent of model construction and timing, producing paired examples. Standard/Block execution order is randomized within each seed using a fixed scheduling RNG; all primary pairs precede exploratory controls.

Real-data profiling selected microbatch four and froze 3,175 optimizer updates per run, with effective batch sixteen (50,800 sampled examples per run). The fixed cost rule projects 14.400 training hours across all eighteen runs. On the RTX 5070 Laptop GPU with PyTorch 2.14.1+cu130, median update times were 0.7760/0.7913/0.8914/0.8471 seconds for Standard/Block/Full/Uniform, respectively. Each median uses sixteen updates after two warm-ups; these short measurements set the horizon rather than establish a throughput ranking. The profile consumed 71.144 seconds; the previous execution checks consumed 2.920 seconds. The source-linked frozen artifact retains all timings, overhead measurements, hardware identity, sample order, and the portable data fingerprint[16].

All variants use AdamW with learning rate 10^{-4}, weight decay 10^{-4}, $(\beta_1, \beta_2) = (0.9, 0.999)$, and $\epsilon = 10^{-8}$. We clip the accumulated gradient norm to 1, use effective batch size 16, BF16 autocast with FP32 loss reduction, 5% linear warmup, and cosine decay to 10^{-6}. A common microbatch is selected from 4, 2, and 1 by memory feasibility. Dropout and stochastic depth remain zero.

The 24 GPU-hour cap allocates 2 hours to profiling/validation, 18 to training, 2 to evaluation, and 2 to contingencies. An exclusive persistent ledger includes failed GPU attempts and wall time spent on IO inside GPU phases. CPU data preparation is outside the GPU allocation. Profiling measures actual optimizer updates for all variants, validation, and checkpoint serialization. A single update count is chosen using measured 90th-percentile update costs, amortized overhead, run multiplicities, and a prespecified 20% safety margin. This count is published before confirmatory training. The final fixed-update checkpoint defines the endpoint; validation scores never select a better checkpoint.

3.4 Metrics and statistical analysis

For standardized target y and prediction $\hat{y}$, field c, and forecast step t,

$$m_{tc} = \frac{\langle (\hat{y}_{tc} - y_{tc})^2 \rangle_{x,y}}{\langle y_{tc}^2 \rangle_{x,y} + 10^{-6}}. \tag{3}$$

For horizon H, errors average equally over active fields and steps $1, \ldots, H$, then predetermined windows within a trajectory, then trajectories within each dataset. The target second moment is

computed in training-standardized coordinates; this definition is not a claim to reproduce every normalization convention used in The Well benchmarks.

Let $E^B_{s,d}$ and $E^S_{s,d}$ denote Block and Standard 12-step dataset errors for seed s. The independent paired effect is

$$D_s = \tfrac{1}{2}\sum_{d=1}^{2} \log(E^B_{s,d}/E^S_{s,d}), \qquad I = 1 - \exp(\overline{D}). \tag{4}$$

We report all six paired effects, a Student-t interval for $\overline{D}$ transformed to I, and an exact two-sided test enumerating all 2^6 sign flips. The interval assumes approximately normal independent seed effects; the randomization test requires within-pair exchangeability under its null. Frames and windows are correlated observations, not independent replicates. Inference is conditional on the fixed test trajectories. With six pairs, the minimum two-sided exact p-value is $2/64 = 0.03125$. Undefined/divergent trajectory errors propagate to dataset estimates; incomplete primary pairs produce an undefined primary analysis rather than a reduced successful-only sample.

Prospective power sensitivity uses assumed Gaussian paired log ratios with standard deviations 0.02, 0.05, 0.10, and 0.20, and improvements of 5%, 10%, and 20%. It includes noncentral-t power and 100,000 independent simulated exact tests per condition, with simulation RNG seed 20261025. These are assumptions, not estimated post-hoc power.

3.5 Secondary outcomes and interventions

Secondary outcomes include one-step and 24-step NMSE, persistence, per-dataset errors, throughput, peak memory, gradient norms, and depth-routing weights. Physical diagnostics include radiative negative-pressure/density fractions and convection velocity-divergence residuals compared with coarsened ground truth using stored physical grid spacing. They do not assert exact conservation or solver equivalence.

On the first four test trajectories per dataset in manifest order, trained Block models are re-evaluated with uniform routing weights and with a fixed-RNG depth permutation of learned weights. These are within-model interventions, not separately retrained models or proof that a mechanism mediates all architecture differences. Their sources and parameters are unchanged; only routing weights are intervened upon. Before test evaluation, an implementation audit replaced the intervention's pointwise FP32 reduction with the same flattened contraction used by ordinary FLA routing. This prevents an additional arithmetic change when weights are held fixed. Exploratory comparisons are reported descriptively without converting secondary tests into a replacement primary endpoint.

4 Results

At this report revision, 18 of 18 planned training runs are complete. Block Attention Residuals reduce the primary error endpoint by 24.52% relative to GPhyT without Attention Residuals, with 95% confidence interval [11.56, 35.58]% and exact paired p-value 0.03125. These estimates condition on the selected datasets and fixed evaluation windows. The interval supports the prespecified minimum 5% improvement. Across the available ordinary evaluations, 0/1368 forecast windows have undefined twelve-step error, and 0/1368 have undefined twenty-four-step error. These descriptive counts include persistence and repeated model evaluations; they are not independent replicates or an inferential sample size. Per-run counts are preserved in rollout_stability.csv. For a 5% improvement, exact-test simulated power is 0.37124 at assumed log-ratio standard deviation

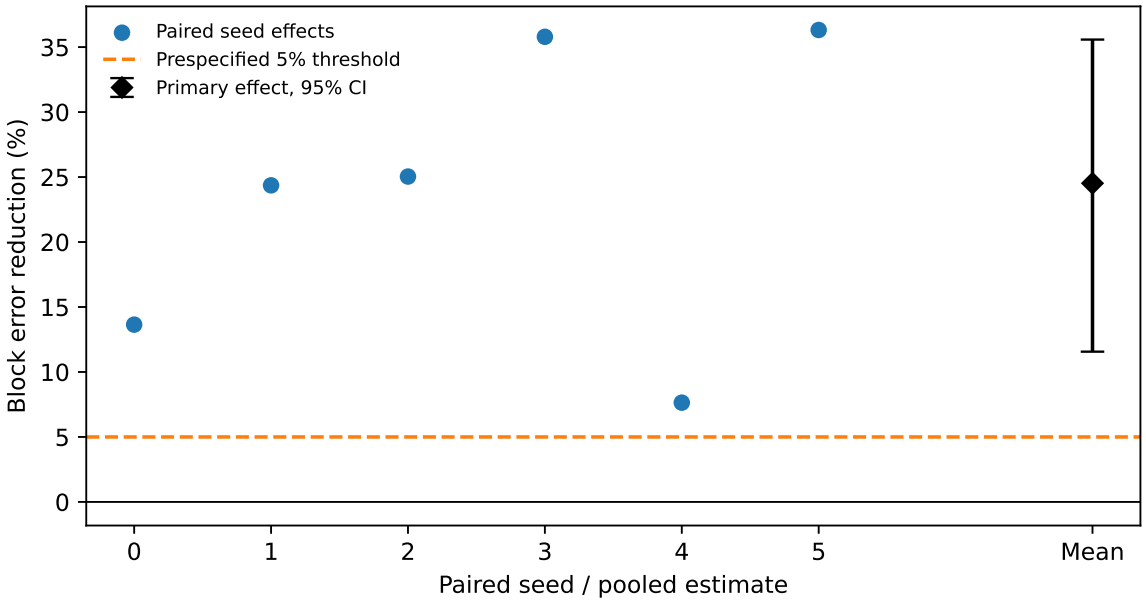


Figure 2: All six paired seed effects and the primary transformed t confidence interval. The dashed line is the prespecified practical threshold.

Model	Convection	Radiative
GPhyT without Attention Residuals	39284.2	0.392769
GPhyT + Block AttnRes	30006	0.289174

Table 1: Primary baseline comparison: arithmetic mean twelve-step NMSE across the same six seeds (lower is better). Both models train from scratch with identical shared-parameter initialization, paired sampled examples, optimizer schedule, update count, and effective batch size. Compute cost differs by architecture. The headline relative reduction uses paired dataset log-error ratios, not the ratio of these arithmetic means.

0.05 and 0.11563 at 0.10 (100,000 draws per condition). All sensitivity assumptions and Monte Carlo errors are published.

Dataset	Model	Finite/planned	1 step	12 step	24 step
Convection	No AttnRes	6/6	334.8	3.928e+04	4.561e+04
Convection	Block	6/6	181.8	3.001e+04	4.076e+04
Convection	Full	3/3	93.01	1.84e+04	3.126e+04
Convection	Uniform	3/3	200.6	2.499e+04	2.898e+04
Convection	Persistence	1/1	0.04142	0.2174	0.3296
Radiative	No AttnRes	6/6	0.1422	0.3928	0.7028
Radiative	Block	6/6	0.1332	0.2892	0.427
Radiative	Full	3/3	0.1381	0.3174	0.4357
Radiative	Uniform	3/3	0.1446	0.3473	0.5948
Radiative	Persistence	1/1	0.1529	0.419	0.5403

Table 2: Arithmetic mean seed NMSE for each fixed horizon. No AttnRes denotes GPhyT without Attention Residuals. A horizon averages all forecast steps up to that horizon. Architecture means require every planned seed to have a finite endpoint. Finite/planned counts refer to twelve steps; persistence is a deterministic baseline, not a one-seed uncertainty estimate. Complete descriptive seed SDs and t intervals are in accuracy_summary.csv.

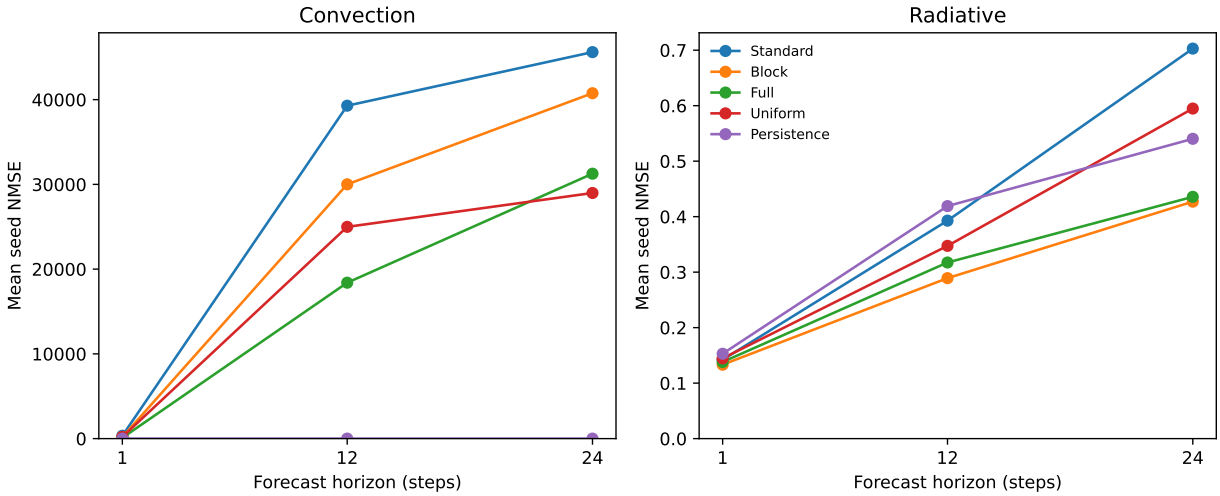


Figure 3: Fixed-horizon trajectory-averaged errors for both datasets. Missing values remain missing.

Run	Updates	Wall seconds	Examples/s	Peak MiB
block_seed0	3175	2473.97	20.68	365.88
block_seed1	3175	2011.53	25.65	365.88
block_seed2	3175	2639.13	19.30	365.88
block_seed3	3175	2547.29	20.36	365.88
block_seed4	3175	2522.62	20.63	365.88
block_seed5	3175	2568.18	20.49	365.88
full_seed0	3175	2966.65	18.11	546.00
full_seed1	3175	2829.46	18.36	546.00
full_seed2	3175	2473.04	20.80	546.00
standard_seed0	3175	2191.09	26.75	278.46
standard_seed1	3175	1282.01	41.12	278.46
standard_seed2	3175	1510.54	33.42	278.46
standard_seed3	3175	1459.29	35.46	278.46
standard_seed4	3175	1488.47	35.03	278.46
standard_seed5	3175	1490.51	34.76	278.46
uniform_block_seed0	3175	2655.30	20.01	345.86
uniform_block_seed1	3175	2470.89	21.12	345.86
uniform_block_seed2	3175	2238.16	23.09	345.86

Table 3: Per-run cost, including failed or incomplete runs. Examples/s uses the median synchronized update time; wall seconds also include validation and checkpointing. Peak memory is PyTorch allocated device memory.

Model / routing control	Convection	Radiative
Block: original	17550.8	0.297138
Block: uniform intervention	42767.8	2.16975
Block: shuffled (random) intervention	40924	2.81369

Table 4: Absolute twelve-step NMSE on the matched diagnostic subset: first four test trajectories per dataset, averaged across all six trained Block seeds. The original reference is evaluated on exactly this subset, not the full test population in the baseline tables. Uniform replaces routing weights by equal weights; shuffled is the random control, permuting learned depth-weight assignments with RNG seed 20261025. Neither intervention is retrained. Seed counts, SDs, and descriptive intervals are in intervention_accuracy_summary.csv.

Dataset	Routing	Finite/planned	Mean error ratio	Seed SD
Convection	shuffle	6/6	2.63	1.245
Convection	uniform	6/6	2.565	0.7757
Radiative	shuffle	6/6	9.524	5.603
Radiative	uniform	6/6	7.49	6.105

Table 5: Within-model twelve-step routing interventions on the same first four test trajectories per dataset. Values are intervened/original NMSE ratios, averaged over trained Block seeds. These are exploratory diagnostics, not independently trained controls or evidence of a unique mechanism.

Dataset	Diagnostic	Standard	Block	Persistence
Convection	Divergence error RMS	4.13	2.82	0.228
Convection	Predicted divergence RMS	4.13	2.81	0.165
Convection	Target divergence RMS	0.175	0.175	0.175
Radiative	Negative density fraction	0.26	0.273	0
Radiative	Negative pressure fraction	0	0	0

Table 6: Secondary physical diagnostics averaged over all 24 forecast steps, then windows, trajectories, and complete planned seed sets. Divergence uses the stored physical coordinates and coarsened target; positivity rates are fractions. These diagnostics do not establish exact conservation. All architectures and interventions are retained in physical_diagnostics.csv.

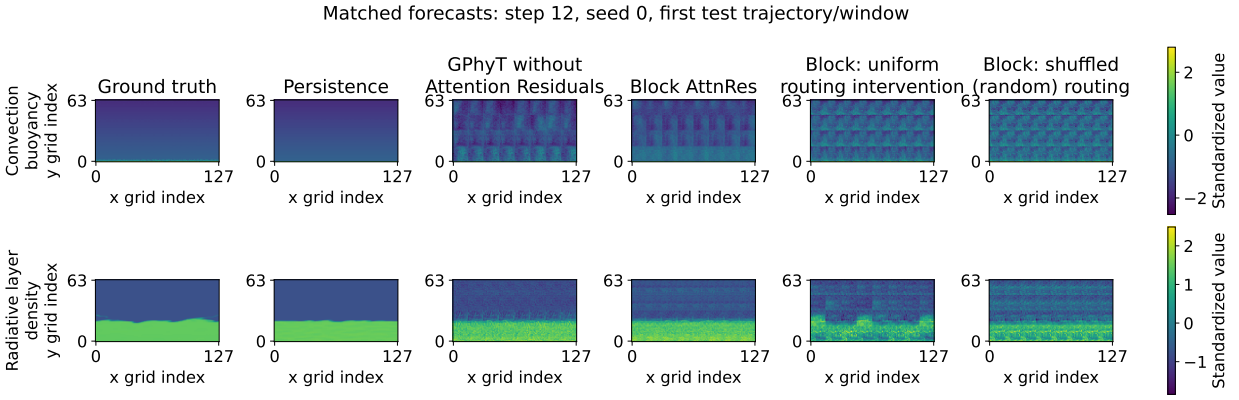


Figure 4: Baselines, new method, and controls on the same first test trajectory/window at step twelve: persistence, GPhyT without Attention Residuals, ordinary Block, and Block with uniform or randomly shuffled depth weights. Interventions use the same trained seed-zero Block checkpoint without retraining; shuffling preserves learned weight magnitudes but permutes source assignments with fixed RNG seed 20261025. Persistence repeats the last observed context frame. Rows show convection buoyancy and radiative density with one shared, unclipped color scale per row in training-standardized units. This post-result presentation reuses fixed cases and saved intervention forecasts; it introduces no new endpoint or GPU inference.

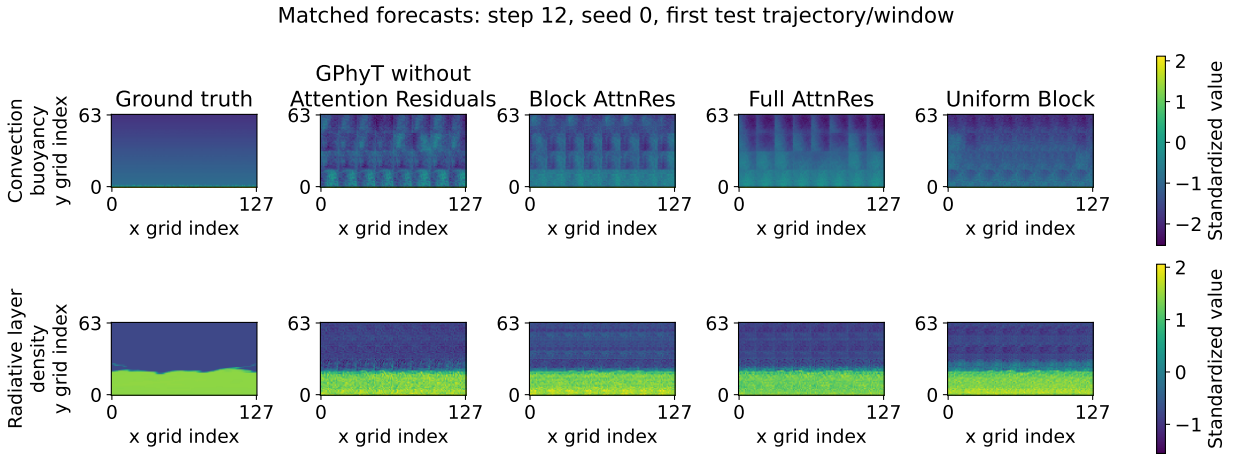


Figure 5: Ground truth and all four trained architectures on identical held-out cases: seed zero, first test trajectory and first fixed window, forecast step twelve. Each row uses one common color range spanning truth and all predictions, without clipping, in training-standardized units; magenta would mark nonfinite values. Convection shows buoyancy (not temperature); radiative flow shows density. These illustrative fields do not replace the full paired evaluation. The all-model layout was added after results at the user's request, reusing the already fixed preview cases.

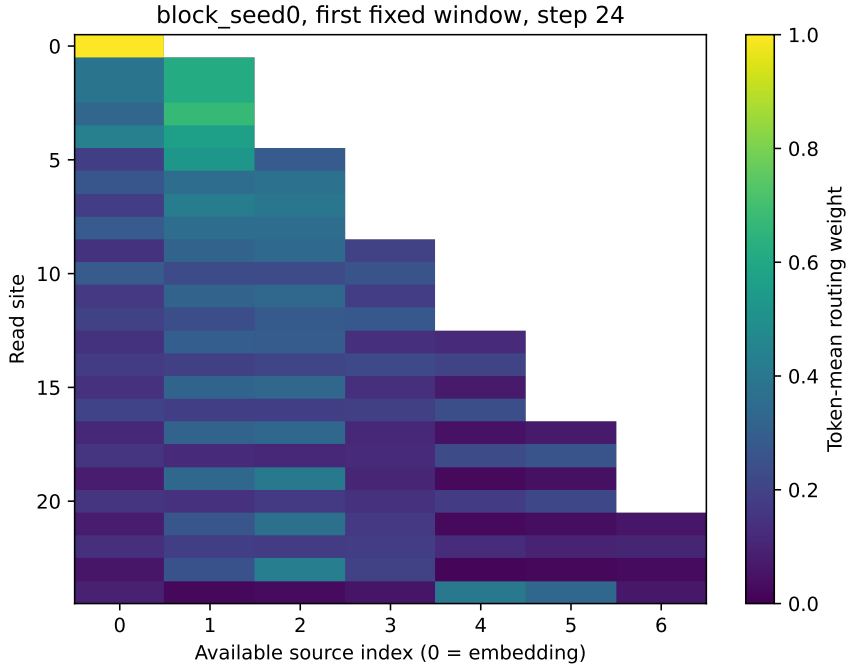


Figure 6: Routing in the prespecified seed-zero, first-trajectory, first-window case. White cells represent unavailable or undefined weights; a heatmap alone does not establish causal importance.

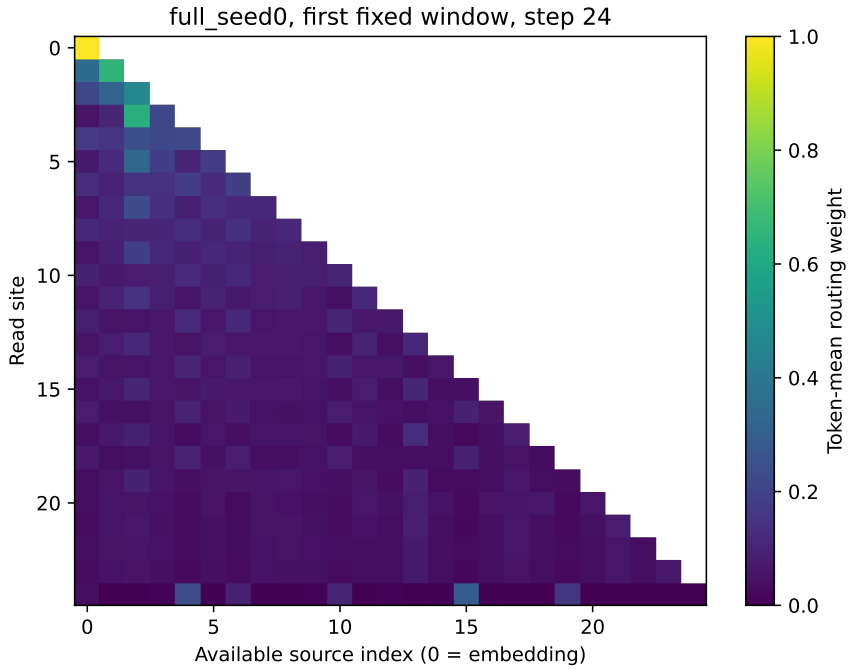


Figure 7: Routing in the prespecified seed-zero, first-trajectory, first-window case. White cells represent unavailable or undefined weights; a heatmap alone does not establish causal importance.

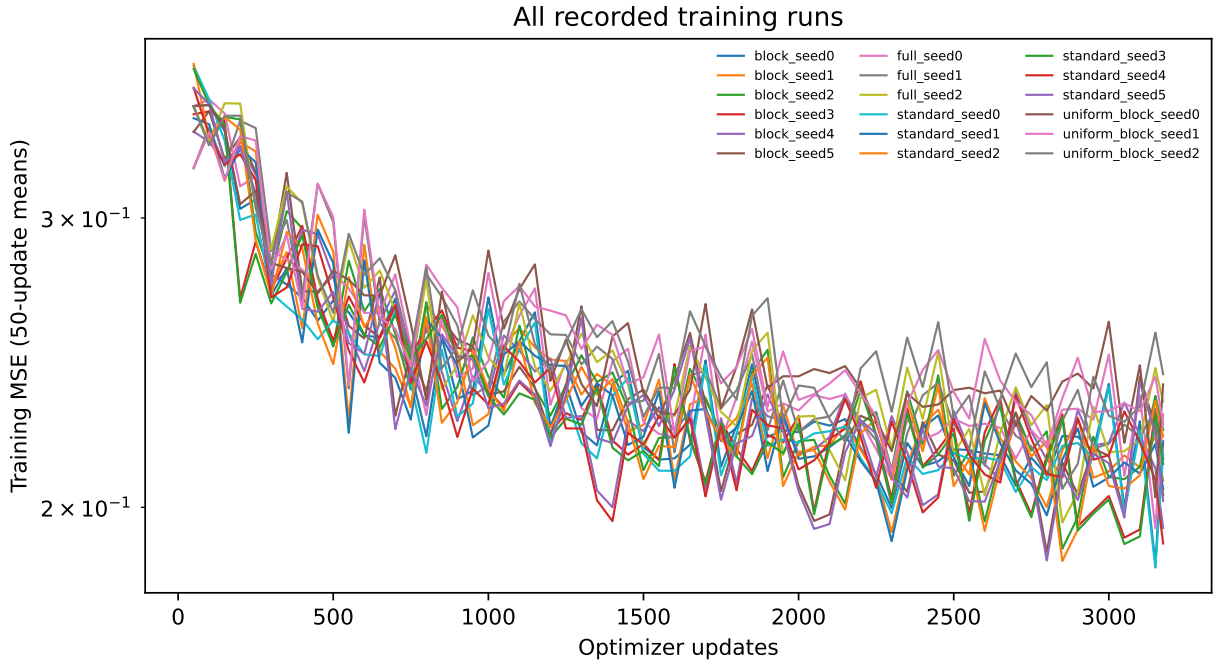


Figure 8: All recorded runs with fixed fifty-update bins. Full update records accompany the checkpoints. Supervised MSE is a training objective, not a reinforcement-learning reward.

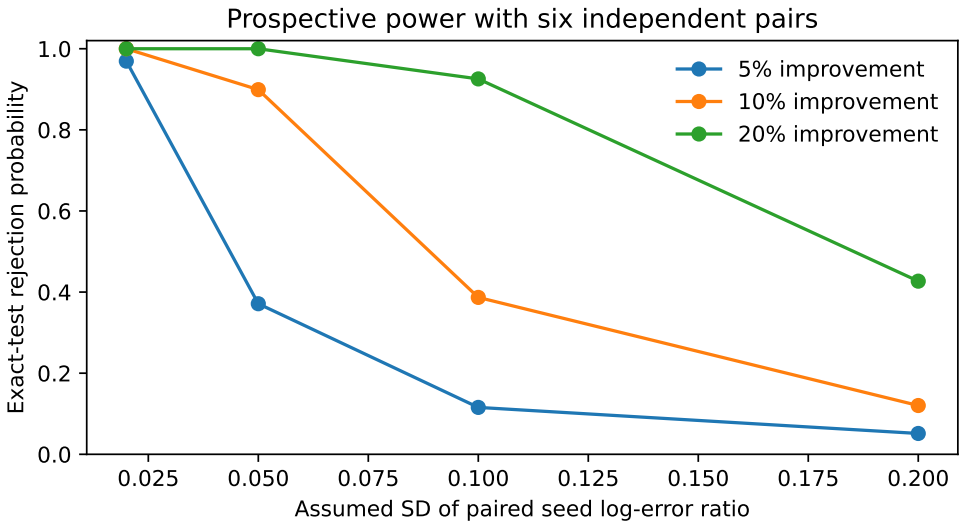


Figure 9: Prospective power under explicitly assumed independent Gaussian seed effects. This is a sensitivity calculation, not observed prediction accuracy.

4.1 Post-hoc initialization-only baselines

The additional user-requested zero-update evaluation completed 18/18 planned controls. These controls were specified after observing the trained-model results and do not alter the original confirmatory comparison. The models were reconstructed with the original constructor, configuration, and seeds, without loading trained weights or applying optimizer updates. Parameter and buffer hashes were checked before and after inference.

Dataset	Model	Finite/planned	Initialized	Trained
Convection	GPhyT without AttnRes	6/6	350742	39284.2
Convection	Block AttnRes	6/6	8173.5	30006
Convection	Full AttnRes	3/3	1332	18397.8
Convection	Uniform Block	3/3	7780.76	24987
Radiative	GPhyT without AttnRes	6/6	7.02088	0.392769
Radiative	Block AttnRes	6/6	0.572704	0.289174
Radiative	Full AttnRes	3/3	0.413095	0.317353
Radiative	Uniform Block	3/3	0.524838	0.347345

Table 7: Twelve-step NMSE for initialized models (zero updates) and their trained counterparts (3,175 updates), on the same complete test population. Standard/Block use seeds 0–5; Full/Uniform use seeds 0–2. Finite/planned counts refer to initialized endpoints; incomplete or divergent seed sets remain undefined. These are descriptive post-hoc controls, not additional confirmatory tests. All horizons and seed uncertainty are in reports/untrained/comparison.csv.

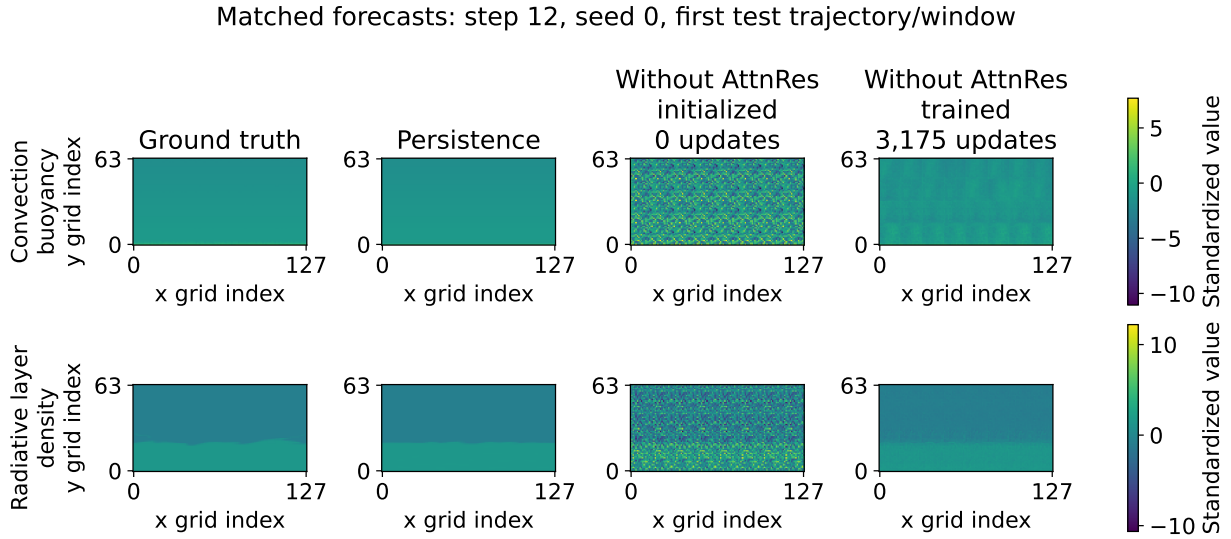


Figure 10: Initialized versus trained GPhyT without AttnRes: ground truth, persistence, zero-update model, and final trained checkpoint. Both rows use the same seed-zero first test trajectory/window at forecast step twelve, with one common unclipped color range per row. Top: convection buoyancy; bottom: radiative density. Magenta denotes nonfinite values. These fixed illustrative cases do not replace evaluation over all trajectories and seeds.

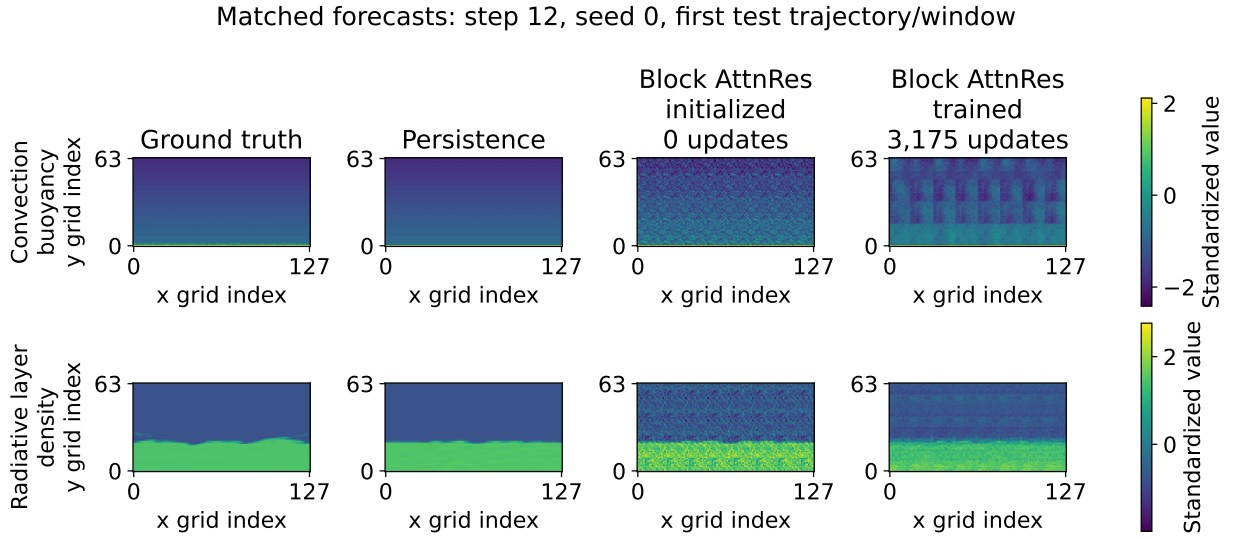


Figure 11: Initialized versus trained Block AttnRes: ground truth, persistence, zero-update model, and final trained checkpoint. Both rows use the same seed-zero first test trajectory/window at forecast step twelve, with one common unclipped color range per row. Top: convection buoyancy; bottom: radiative density. Magenta denotes nonfinite values. These fixed illustrative cases do not replace evaluation over all trajectories and seeds.

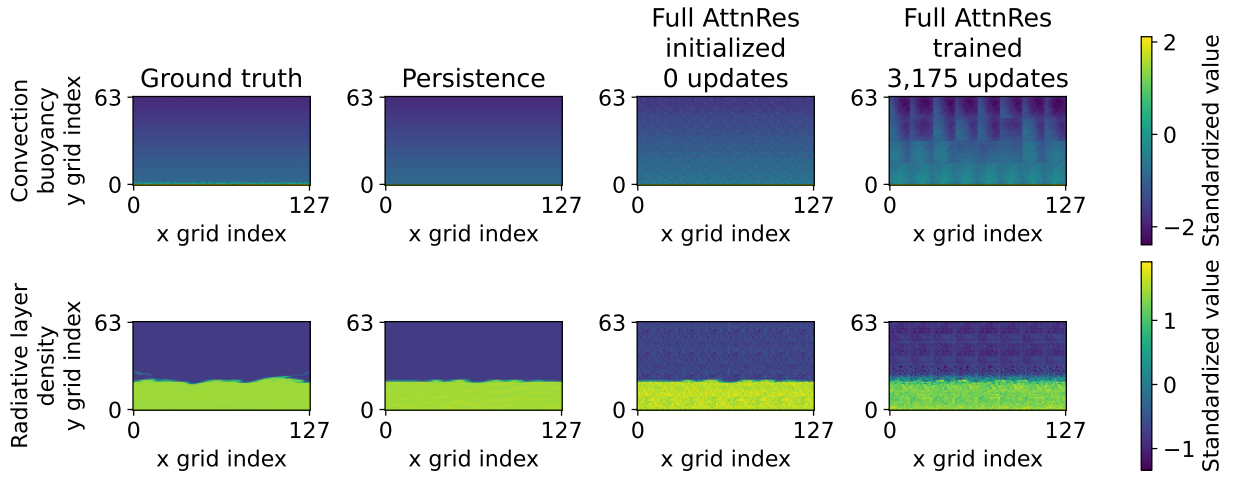


Figure 12: Initialized versus trained Full AttnRes: ground truth, persistence, zero-update model, and final trained checkpoint. Both rows use the same seed-zero first test trajectory/window at forecast step twelve, with one common unclipped color range per row. Top: convection buoyancy; bottom: radiative density. Magenta denotes nonfinite values. These fixed illustrative cases do not replace evaluation over all trajectories and seeds.

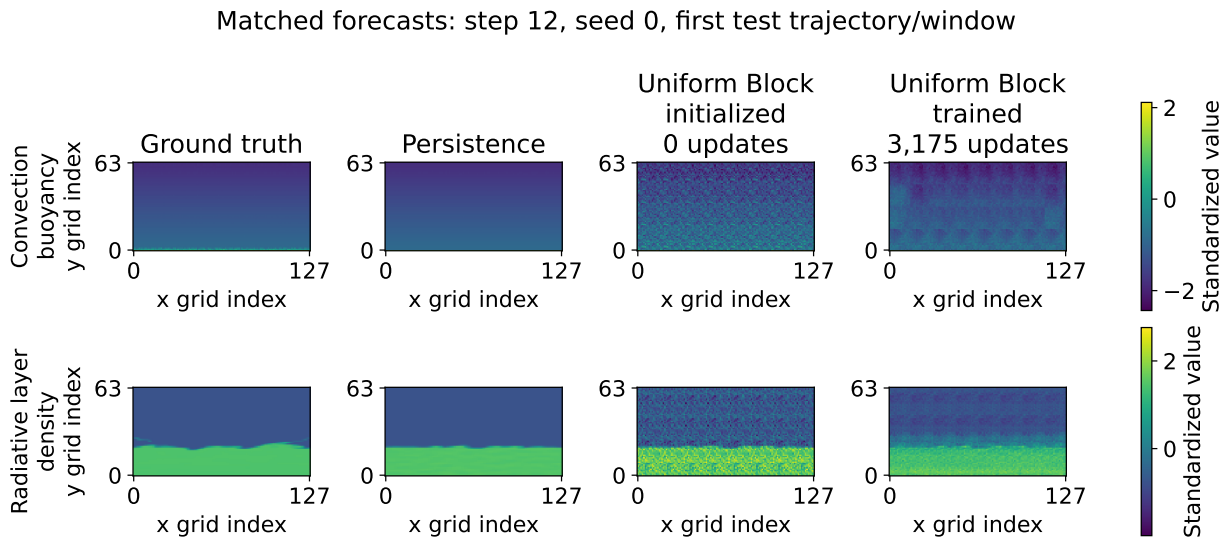


Figure 13: Initialized versus trained Uniform Block: ground truth, persistence, zero-update model, and final trained checkpoint. Both rows use the same seed-zero first test trajectory/window at forecast step twelve, with one common unclipped color range per row. Top: convection buoyancy; bottom: radiative density. Magenta denotes nonfinite values. These fixed illustrative cases do not replace evaluation over all trajectories and seeds.

5 Discussion, claims, and implications

The experiment separates a practical architecture comparison from a mechanistic question. A Standard–Block difference estimates the total effect of replacing residual composition under the matched training protocol. The Uniform Block control helps assess whether learned selection contributes beyond the change to normalized averaging, but its three-seed sample is exploratory. Interventions on trained Block weights examine immediate routing dependence while retaining the learned backbone; they are not interchangeable with training a uniform-routing model from scratch.

All six paired seed log ratios favor Block over Standard. The estimated 24.52% reduction and interval [11.56,35.58]% support the prespecified 5% threshold under the stated conditional inference assumptions. The exact $p = 0.03125$ is the smallest attainable two-sided value with six pairs, not evidence of high-resolution tail estimation. This primary result concerns the geometric combination of dataset error ratios; it is not a 24.52% reduction in every field or dataset.

5.1 Relative gains and absolute forecasting failures

At twelve steps, arithmetic mean seed NMSE is 39,284 for Standard and 30,006 for Block on convection, compared with 0.2174 for persistence. Every trained variant loses decisively to persistence there, including the exploratory controls. On the radiative layer the corresponding Standard, Block, and persistence errors are 0.3928, 0.2892, and 0.4190. At twenty-four steps, Block's radiative error is 0.4270 versus Standard's 0.7028 and persistence's 0.5403. Relative improvement over a weak learned baseline must therefore be distinguished from a useful physics surrogate.

A post-hoc decomposition of the already saved, prespecified seed-zero first convection example clarifies one metric sensitivity without changing the endpoint. For Standard, the first-twelve-step per-field NMSE values are 0.502 for pressure, 0.0394 for buoyancy, 15,707 for horizontal velocity, and 34,062 for vertical velocity. The minimum standardized velocity target energies in this window are 1.19×10^{-7} and 2.67×10^{-8}, below the fixed 10^{-6} stabilizer. Consequently, modest standardized velocity errors receive very large NMSE weights. The independent decomposition exactly reproduces that saved window's aggregate[16]. This single fixed example explains an observed contribution, not every large convection error. Epsilon, windows, and field weights were not changed after observing it.

All evaluated forecasts remain finite through 24 steps. Finite arithmetic does not establish physical stability: the mean convection divergence error is 4.134 for Standard, 2.823 for Block, and 0.228 for persistence. Radiative negative-density fractions are 26.04% for Standard and 27.30% for Block, versus zero for persistence, averaged over all 24 forecast steps and the specified trajectories. Thus the NMSE gain does not improve this positivity diagnostic. No negative-pressure cells were recorded in these ordinary evaluations. No clipping or positivity repair is applied.

5.2 Exploratory controls and routing dependence

Using only matched seeds 0–2 and the same equal-dataset log weighting, Full's geometric error ratio to Block is 0.8349, while Uniform Block's is 1.0238. The three Uniform/Block paired log ratios include both signs. These exploratory observations do not establish that learning depth weights is necessary for the primary Standard–Block gain; normalized source aggregation is also changed by that comparison.

On the fixed diagnostic subset, replacing trained Block routing with uniform weights increases twelve-step error by mean factors 2.565 on convection and 7.490 on radiative flow. Shuffling depth weights gives factors 2.630 and 9.524. Every tested seed's factor exceeds one for each of these

comparisons. The precision-matched interventions demonstrate dependence on routing in those trained models and selected trajectories. They do not identify a unique mediator of the training-time architecture effect, and the subset's manifest ordering does not cover all parameter settings.

5.3 What the initialization-only controls reveal

The later user-requested controls compare reconstructed initial states and final checkpoints for each original architecture/seed identity on the same complete test population. Training reduces twelve-step radiative-flow error for all eighteen identities. On convection, Standard improves for all six seeds, but Block, Full, and Uniform Block worsen in all six, three, and three seeds, respectively. Their trained-to-initialized ratios of arithmetic mean convection NMSE are 3.671, 13.812, and 3.211. The corresponding Block means are 8,173.50 at initialization and 30,005.97 after training; Full's are 1,332.00 and 18,397.80. Thus final-checkpoint superiority to Standard does not establish improvement over these architectures' own initial states on convection. Large architecture differences already exist before learning. These are descriptive post-hoc findings, not new confirmatory tests; they neither change the endpoint nor identify a unique cause of degradation[16].

Block and Uniform Block have identical initialized tensor hashes and bitwise-identical saved forecasts for their shared seeds 0–2. Their reported overall initialized means differ because Block also includes seeds 3–5. Identical initial states must not be counted as independent evidence for a learned-routing benefit. All four architectures, initialized or trained, still lose to persistence on convection under the fixed metric. The held-out initial-state overlap and low-energy metric sensitivity apply equally to these new controls.

5.4 Measured cost

All eighteen runs finished 3,175 updates, with no failed or omitted training runs. Training consumed 11.068 GPU-phase hours, below its 18-hour allocation and the pretraining 14.4-hour projection. Mean per-run wall times are 1,570 seconds for Standard, 2,460 for Block, 2,756 for Full, and 2,455 for Uniform. Mean run-level throughputs are respectively 34.42, 21.19, 19.09, and 21.41 examples per second; PyTorch peak allocated memory is 278.46, 365.88, 546.00, and 345.86 MiB. These are measurements on one laptop GPU, including varying host conditions, not universal kernel benchmarks. Equal optimizer updates match data exposure while leaving compute cost unequal.

6 Limitations

Six independent pairs constrain both power and the precision of variance estimates. Three-seed controls cannot support broad systematic claims. The confidence interval is conditional on selected trajectories and its normality approximation; no claim of coverage under arbitrary effect distributions is made. Matching optimizer updates does not match wall-clock compute because routing architectures have different costs. Conversely, equal compute would change the data exposure and is not the primary comparison specified here.

The routing-intervention subset is chosen by manifest order and is not balanced over every parameter setting. Its immediate within-model effects cannot establish that routing explains behavior across all held-out regimes. Descriptive secondary intervals are unadjusted and do not replace the preregistered primary test.

All test trajectories share their initial state with training at a different physical parameter setting, although neither complete trajectories nor their first four prepared frames are exact cross-split duplicates. The official split therefore does not establish generalization to entirely unseen initial

conditions. Initial-state reuse and related simulation families also limit extrapolation beyond this fixed benchmark; the seed-level inference is conditional on it.

Coarsening removes unresolved spatial structure and changes numerical divergence residuals. Training-standardized NMSE can strongly weight low-energy targets despite the fixed stabilizer. The model learns a mixture of temporal strides without an explicit stride token, preserving the upstream forward interface; this may limit temporal identifiability. A 12-step endpoint is not a guarantee of asymptotic dynamical stability. GPU execution can vary across hardware and library versions even with deterministic algorithms requested[14]. All these limitations apply regardless of the sign of the observed effect.

7 Conclusion

Block Attention Residuals reduces the prespecified paired error endpoint relative to Standard GPhyT in this completed 18-run study. That evidence is conditional on two selected official splits, 128-by-64 resolution, GPhyT-S, and 3,175 updates per run. It coexists with severe convection failure against persistence and frequent unphysical radiative density predictions. The controls leave the incremental benefit of learning routing over fixed averaging unresolved, while within-model interventions show routing dependence on their fixed subset. These findings justify a bounded architecture result and targeted future hypotheses, not claims of solver reliability or general physics generalization. The additional initialization-only baselines also show convection degradation after training for all three routing variants. This failure is visible despite their favorable final-checkpoint comparison with Standard.

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A Artifacts and audit trail

The public repository[16] contains the locked Python 3.13 environment, retained upstream licenses, source hashes, tests, study configuration, preregistered protocol, prospective power tables, and phase-based CLI. Large data and checkpoints live in a configured project-owned directory. The model release uses tensor-only safetensors weights; local optimizer-resume files are trusted artifacts and are not loaded from arbitrary third parties.

Reproduction proceeds through preparation, CPU tests, prospective power, budgeted GPU verification, profiling, publication of the frozen update count, training, evaluation, analysis, and reporting. The README is the authoritative command reference. Checkpoints preserve optimizer moments, completed update number, CPU/CUDA RNG states, and the independent data-sampling RNG. The learning-rate schedule is reconstructed from the saved update and frozen horizon. The published manifest preserves official split membership and training-only normalization.

B Verification scope

Tests compare the FLA operator with the stated equations and gradients, test zero-query averaging, causal source counts, final aggregation, embedding inclusion, forward-call reset, and preservation of the upstream standard path. Data checks cover field mapping, split duplication, normalization isolation, and fixed forecast origins. Forecasting is structurally separated from future targets. Checkpoint tests restore optimization and sample state. Budget rejection paths retain failed attempts. The dashboard and final PDF require visual verification in addition to numerical tests.

Additional regressions require bitwise equality when uniform or shuffled interventions leave routing weights unchanged, both with and without BF16 autocast. Nonuniform identity-shuffle probes cover learned routing. Separate output and gradient comparisons protect the ordinary FLA training path. All sixteen cases passed on CPU and, after training, on CUDA under the original validation allowance, before held-out evaluation. The dated architecture audit records the failing and passing checks. The initial release CPU suite passed 44 tests. Six additional CPU tests check matched targets, active fields, forecast dimensions, missing models, and nonfinite predictions in the saved field comparisons. Five further tests check matched intervention score reduction, missing seeds, inconsistent references, persistence independence from future frames, and target agreement for random-control previews; these passed with the existing reporting checks (16 tests in that focused run). An independent CPU audit reconstructs the six primary effects, enumerates all 64 sign assignments, and reproduces the confidence interval from the raw saved endpoints[16].

All eighteen final checkpoints and complete training logs have SHA-256 receipts. The four public variant repositories are grouped in the BurnyCoder GPhyT Attention Residuals collection. The release verification selects Standard seed zero by frozen execution order, checks downloaded bytes against the original checkpoint, and compares all three horizon errors on the first test trajectory of each dataset. Publication evidence and verification metrics are stored in the public release receipt.

C Reporting policy

The later initialization-only extension reconstructs all eighteen original architecture/seed identities at zero optimizer updates. Its separate published protocol and implementation precede these additional evaluations; the extension is post-hoc relative to the original results. No historical step-zero checkpoint was saved during training, so the released initial weights are reconstructions using the original constructor, frozen configuration, and seed. No trained checkpoint is loaded and no optimizer is created. Complete parameter/buffer hashes must match before and after inference; shared-backbone hashes must agree within each seed. The same causal evaluator, complete test population, training-only normalization, and fixed windows are reused. Initial weights and evaluation metadata are published under `untrained/seedN/` inside the same four model repositories, with a separate zero-update configuration. The existing evaluation allowance and total GPU cap also cover this extension. Its scores, descriptive seed uncertainty, per-run reports, and fixed field examples are isolated under `reports/untrained/`, leaving the original primary analysis unchanged[16].

Every planned experiment retains a report even if omitted or failed. Raw per-trajectory metrics, training logs, runtime costs, configuration hashes, and source versions support derived tables. Post-hoc analyses must be labeled separately from the primary endpoint. No reinforcement-learning reward is defined for this supervised regression study, so no reward curve is reported. Public model cards must state the same data, normalization, compute, and scientific limits as this manuscript.

AI authorship and human revision statement

Codex wrote this codebase and paper. A human provided the exact specification that Codex executed. Human cleanup and rewriting are in progress.